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AGI ALPHA FLYWHEEL

— When Capability Creates Opportunity

Transformational Intelligence, Experience-Trained Specialists and the Expanding Frontier of Profitable Work

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Keep the measurement.
Change the frame.

An executable instrument for carrying useful scientific meaning into fresh work—and a transparent test of when retention pays.

48/48
Fresh compatible missions
Competent analytical and identical-library controls also pass.

12/12
Meaningful refusals
Unsupported meanings and malformed inputs stay outside scope.

17.31 min
Conditional retention threshold
Four realized uses; assumed costs. No time saving measured.

FIND YOUR ALPHA. BUILD YOUR SUCCESSOR Ω.

Actual public data and executed software. Simulated commerce. No claimed commercial, graph or model-quality gain.

A measurement worth carrying forward

A laboratory has already paid to acquire an image and delineate its objects. A collaborator needs those objects in another coordinate frame. The useful result is an answer whose meaning survives the move: which supplied regions were measured, how much native physical space they occupy, and where their centers lie under a declared geometric map. A converted file or attractive rendering cannot establish that result by itself.

This study constructs and executes a small instrument for that purpose. It binds a measurement contract to exact source bytes, computes the requested quantities, rejects unsupported meanings, and packages executable behavior for another recipient. After source freeze, the recovered instrument accepts **48 of 48 compatible fresh missions** on a separate public microscopy study and correctly refuses **12 of 12 incompatible or malformed missions**. The artifact is actually executed in new Python processes; transfer is more than retrieval of a document. [E1–E4]

The strongest ordinary alternative matters. A competent analytical workflow accepts the same 48 missions, and an ordinary directory with identical artifact contents produces identical scientific outputs. A separate, small hosted-model trial yields **12/12 accepted outcomes for U, E and E+**. Thus the study establishes a usable instrument and fresh executable transfer; it establishes no accepted-quality advantage over the competent recipient, no graph-specific benefit, and no measured complete-cost advantage. [E3, E5]

The commercial study asks what would make such a service worth operating and such a capability worth retaining. Under explicit buyer, effort and failure assumptions, a frozen expert operating policy produces mean simulated economic surplus of **CAD596.86**, comprising a fictional originating-job surplus of CAD220 and CAD376.86 from subsequent work. Actual customers and collections are zero; complete actual research cost and operating profit are unknown. The primary retention policy rejects an unproven increment after a CAD5 screen. With four realized eligible uses, a CAD75 additional retention burden would require more than **17.31 minutes of avoidable CAD65/hour expense per use**. No such saving was observed. [E6]

The connected achievement

Step	What this edition establishes	Evidence boundary
Consequential need	Preserve the meaning of quantitative microscopy across an affine handoff	A narrow service hypothesis; no customer demand observed
Useful realization	Source-bound native and mapped measurements, with explicit refusal	Published label support, not validated biological truth
Fresh recipient use	Exact recovered code succeeds on 48 new affine missions	Two label groups in one image/timepoint
Complete accounting	Executable conditional A/B/C and cash-path model	Simulated transactions; actual costs remain incomplete
Selective retention	Testable break-even rule and a justified primary stop	No demonstrated additional Alpha or compounding

The contribution is an inspectable connection between purpose, construction, commercial conditions and inheritance. The affine mathematics is established. The useful new artifact is the joined contract, bounded implementation, evidence trail and integration with existing lifecycle interfaces. Experience here means retained executable structure developed through tests and correction; proprietary model weights were not trained. “Specialist ASI” remains an ambition.

1. Explore the need before choosing the test

The beneficiary is a research laboratory or imaging core transferring quantitative results between pipelines, instruments or collaborators. Its desired outcome is an accepted measurement report with reproducible source identity and explicit scope. The costly dependency under investigation is a workflow convention: rebuilding a target raster before measuring an object whose requested aggregate quantities can instead be derived directly from source support.

A bounded search considered four materially different candidates. The dossier records 20 search queries and 14 opened primary pages, plus later contrary references; the final source inventory contains 22 entries with retrieval status preserved. The initial discovery limit was extended for three decisive existing-method references, to prevent a false novelty claim. This was enough to select a tractable, consequential test, not a global novelty search. [E7]

Candidate and beneficiary	What already exists	Consequential gap or reason to redirect
Microscopy continuity: laboratories and imaging cores	NGFF metadata, image/label formats, resampling, physical feature extraction, validators	A format-valid export does not by itself qualify the requested quantitative meaning; small real licensed inputs and clear falsifiers are available
Accessible transit contingencies: passengers and operators	GTFS Pathways and OpenTripPlanner already model wheelchair restrictions and unknown accessibility	Fresh outage and physical-access observations may remain missing; live reliability obligations exceed this bounded experiment [17]
Water-system diagnosis: utility analysts	WNTR, sensor-placement methods and fault-aware diagnosis	An extra observation might separate ambiguous hypotheses; a distinct advantage and data redistribution rights were not established [18]
Materials experiment campaigns: formulation teams	NIST TDE and Autonomous Formulation Lab already supply planning and measurement capabilities	No distinct unresolved application gap established within budget; physical validation would dominate [19]

The microscopy choice has accessible real inputs, a meaningful failure mode and an equivalent human-service reference. It also has strong prior art. NGFF describes coordinate systems and transformations; its specification excludes exact transformed/resampled pixel reproducibility. SimpleITK already exposes per-label physical size and centroid, scikit-image exposes spacing-aware region properties, and ITK separates spatial objects from their presentation. These are building blocks and serious comparators. [1–5]

The search therefore did **not** discover affine moment transport as a new algorithm, nor establish that imaging specialists lack it. It identified a purpose-specific packaging and verification question: can a small artifact make the intended quantity explicit, survive a recipient handoff, and refuse changes that invalidate that meaning? The falsifier for a claimed E+ advantage is straightforward: if the competent existing workflow meets the same contract at equal or lower complete cost, the new capability adds no demonstrated advantage to that recipient.

The declared working budgets were a short discovery phase, 35 minutes for initial implementation, 35 for evaluation and 40 for writing, with extensions only for concrete unresolved issues. Source and task commitments bound the actual experiment. These are planning budgets, not measured labor or a claim that every phase finished within its allowance. Failed and revised development attempts remain in the archive. [E1, E7, E8]

2. The transformation witness

The useful purpose is narrower than "preserve the image." It is to report supplied label identities, occupied native physical measure and centroids, and the geometric pushforward of those same regions under an explicit affine map. Display pixels, object topology, fluorescence intensity and biological correctness are separate purposes.

Witness element	Inspectable content
Original purpose	Recover the stated geometric quantities for the supplied non-background label support, with exact source identity, declared units and meaning
Binding restriction	A named common-grid pipeline requires raster resampling before measurement; discretization can change occupied support
Why removable	Counts, first moments and support measure under an affine map have an analytical relation to the source partition; a target raster is not required for these quantities
Replacement	Keep the native partition and metric; transport geometric moments through the declared map; keep native and mapped measures distinct
Executable realization	Frozen `microscopy_transfer.py`, a validated contract, immutable byte binding, machine-readable output and refusal reasons
Falsifying observation	Wrong IDs/counts, measure or centroid outside tolerance; unsupported meaning accepted; source mismatch accepted; or an advantage claim defeated by the competent comparator

This replaces a restriction of the **named pipeline**, not a proven ceiling on all existing methods. A source-space analytical baseline receives the same formulation and permissions. Its success separates the value of the formulation from any claim of better execution or learned discovery. [E1–E3]

Three semantic distinctions preserve the original purpose. First, the number of label identities is not the number of connected components or independently verified biological cells. A single label can have disconnected support. Second, native physical measure remains tied to the source metric. Mapped geometric measure may change under a deformation. A coordinate re-expression alone does not physically enlarge the specimen. Third, an explicit map is supplied input; this study neither estimates nor validates biological registration.

The instrument accepts only missions that make these distinctions explicit. It can refuse an unresolved axis, unsupported meaning or conflicting source identity. It cannot infer that a mathematically valid matrix secretly represents the caller's unintended inverse map. That remaining responsibility belongs to mission preparation and acceptance, and it must be costed in any service trial.

What was supplied, constructed and observed

The source labels, standard numerical libraries, affine formulas, task semantics, allowed transformations and simulation parameters were supplied. The team constructed the bounded implementation, contract checks, test generators, independent numerical oracle, transfer adapter and delivery evidence. A curator selected the confirmation source before inspecting outcomes; the instrument author froze the implementation before that inspection. This is role separation within one AI-assisted team, not external validation. The commercial alternatives were researcher-defined ablations; compatible customers and financing were not autonomously discovered. [E1–E7]

3. Carry the quantity, not an unnecessary raster

Let label k occupy index centers I_k in a d-dimensional source array. Let $D = \mathrm{diag}(s_1, \ldots, s_d)$ contain positive physical spacings, and o the source origin. The contract treats integer index i as a voxel center and its support as $i + [-1/2, 1/2)^d$. For $n_k = |I_k|$,

$$V_k = n_k \prod_{j=1}^{d} s_j, \qquad c_k = o + D\left(\frac{1}{n_k}\sum_{i \in I_k} i\right).$$

For an explicit physical-coordinate affine map $T(x) = Ax + b$, with nonsingular A,

$$V_k^{\mathrm{mapped}} = |\det A|\, V_k, \qquad c_k^{\mathrm{mapped}} = Ac_k + b.$$

These identities follow from affine change of variables and linearity of the first moment. Reflection uses the absolute determinant. Native volume V_k stays in the source metric; the second volume describes the mapped geometric support. In two dimensions, the same formulation gives area. This is established mathematics, implemented here for a precisely bounded handoff. [3–5, E1]

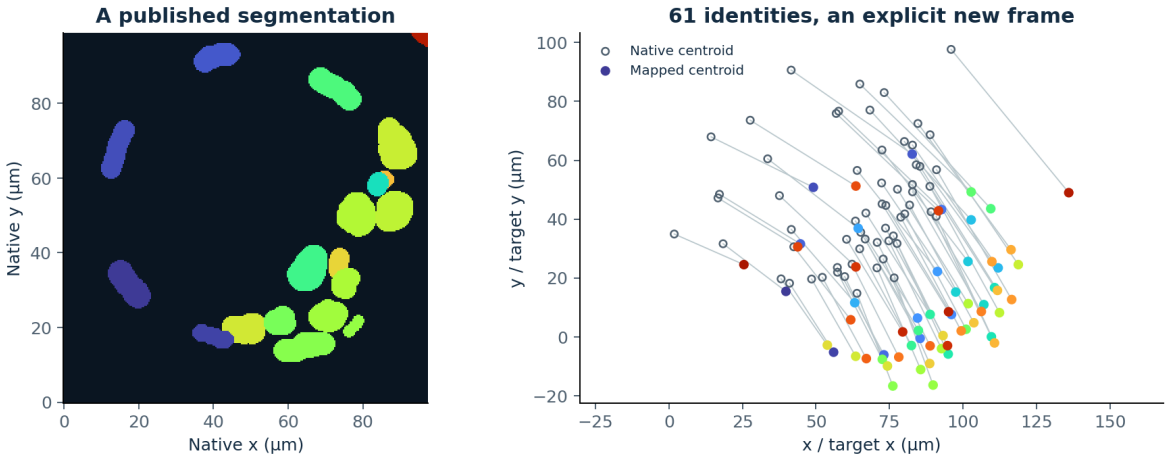


Figure 1: A published source partition and its centroids carried into a new frame. Left: actual IDR0062 label slice. Right: native and mapped centroids from the recorded development result. The transformation is authored; biological validity is not inferred. Source imagery: Blin et al., CC BY 4.0.

The implementation takes a single owned, bounded NPY byte snapshot, hashes those exact bytes, checks the header and payload before loading, and then measures the trusted array. The public CLI performs this binding; the internal array helper assumes trusted input. The bound is one GiB per source. Integer labels may be signed and nonzero, with magnitude at most $2^{53} - 1$ for exact JSON interchange; zero is background. [E1, E8]

Source calibration is axis-aligned. Affine columns follow `source_axes`, rows follow `target_axes`; maps are expressed in physical micrometers. The contract preserves the requested label set and numerical tolerance. It does not silently substitute display resampling, nonlinear mapping, topology, intensity, segmentation correction or clinical interpretation. This limited scope is what makes the instrument auditable and the experiment decisive.

4. A real source, a revealing diagnostic

Development used IDR0062A image 6001240 from the published NesSys study. Its native label array has shape 236 × 275 × 271, spacing 0.5002025531914894 / 0.3603981534640209 / 0.3603981534640209 micrometers, and 61 nonzero label identities. Four labels touch a source boundary; they are not asserted to be complete cells. All 32 compressed chunks and metadata were preserved. The exported array has SHA-256 beginning `c557c80743037400`; its full digest is in the provenance record. [6, E2]

The constructed instrument and conventional analytical reference returned the same label set and measures. Maximum centroid disagreement was 1.42×10^{-14} micrometers. The instrument's recorded numerical work took 3.156 seconds; conventional computation took 0.784 seconds. This is successful numerical realization, with no speed advantage claimed. [E2]

A separate named diagnostic applied SciPy nearest-neighbor resampling to common grids before measurement. It kept all 61 IDs but changed their support measures. At 0.75 micrometers, six regions exceeded 1% discrepancy and the maximum was 13.262%. At 1.5 micrometers, 26 exceeded 1%; at 3 micrometers, 51 did. The analytical method avoids that particular discretization step. [E2]

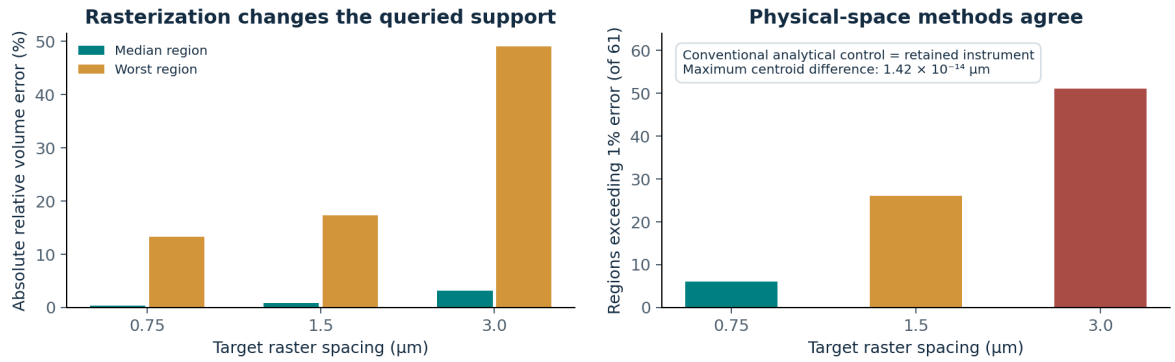


Figure 2: Observed common-grid diagnostic on the development array. Larger grid steps increase discrepancy in the measured support of this partition. This is a named raster pipeline, not an ordinary-user frontier-model baseline. The 1% line is a readable diagnostic; the engineering contract uses a stricter analytical numerical tolerance.

These results do not show that resampling is always inappropriate. It is useful for display and for tasks that genuinely require a target raster. They show that an unnecessary representation change can alter a requested quantity. They also do not show that a competent analyst would choose the deficient route: the strong analytical baseline already avoids it. The transformation is useful at the workflow level; uniqueness, a new market and Operational Alpha remain unproved.

5. Fresh recipient use after a real source freeze

The capability was frozen at 17:40:14 UTC on 27 October 2026. Confirmation tasks had been committed at 17:37:23 UTC. The first confirmation completed at 17:42:57 UTC. The module SHA-256 remained `a04326ce0504bfe3ac290b6cb65ee6f089cc66a607f70b7cc7869099250c77db`. No held result was used to change it. [E1, E3]

Confirmation used a different public study: IDR0052A image 5514375, time index 7 and channel 0, with the published `Cell` and `Chromosomes` label groups. Each array is 31 × 256 × 256 with spacing 0.75 / 0.2516 / 0.2516 micrometers. Each contains supplied label 8. Their occupied native measures are 5,489.85121344 and 865.88406696 cubic micrometers, respectively. These are properties of the published partitions, not new biological findings. [7, E3]

Each group receives 24 authored affine missions: identity, translation, reflection, anisotropic scale, shear and seeded combinations. The 48 missions therefore exercise two groups from one image/timepoint; they are not 48 independent biological specimens. The recipient executes the exact artifact recovered through the existing lifecycle fixture in a new interpreter per mission. The ordinary-library control uses identical bytes and rights in an ordinary directory. [E3, E4]

Configuration	Accepted compatible missions	What the comparison establishes
Recovered executable, new recipient process	**48/48**	Actual reusable execution on fresh missions
Conventional analytical method, with caching	**48/48**	Existing competent methods meet the same contract
Ordinary directory, identical executable	**48/48**	Exact scientific parity; no graph-specific gain
Incompatible or malformed missions	**12/12 correct refusals**	Meaning and input boundaries are enforced in tested cases

The independent reference enumerates every occupied voxel center, transforms those points, and averages them. It derives mapped voxel measure from the scalar triple product of transformed basis vectors. Exact IDs/counts are required; relative measure error must not exceed 10^{-9}, and centroid discrepancy must not exceed 10^{-8} micrometers. Maximum recovered discrepancies were 4.40×10^{-16} and 1.074×10^{-11} micrometers. These are numerical consistency limits, not microscope precision. [E3]

The 12 refusal cases cover invalid spacing, singular or nonfinite maps, source-byte mismatch, nonlinear mapping, intensity, topology, conflated native/mapped volume, an unsupported reverse-direction declaration and an unresolved time axis. The refusal of an explicit reverse-direction field does not establish detection of hidden human intent.

The recipient's 48 external invocations took 11.542 seconds, including startup and CLI work; its internal numerical timer totals 2.200 seconds. Conventional summary preparation plus mapping took 0.333 seconds within the evaluator process. The frozen CLI recomputes source moments per invocation, while the conventional baseline caches its source summaries across maps. These clocks have different deployment overheads, so they are diagnostics, not a matched latency claim. They provide no evidence that the new instrument is faster. [E3]

6. Operational Alpha receives its own test

A useful workflow can improve delivery without new algorithms or trained weights. To distinguish this possibility from retention, the study defines four explicit references. H remains the primary commercial comparator; U, E and E+ separate model operation from the extra retained asset.

Reference	Definition in this edition	Observed or modeled evidence
H	Equivalent human image-analysis service	Sourced rate; task hours and scope match are assumptions
U	Ordinary same-model worker given the complete task and tools	12/12 accepted outcomes
E	Same task and tools plus a frozen short operating rubric	12/12 accepted outcomes
E+	E plus the exact recovered executable	12/12 accepted outcomes

Three fresh hosted workers received the same 12 authored tasks and ten-tool-invocation/ten-minute ceilings. U received every essential requirement, including source identity and quantity definitions. E added a short method: distinguish native/mapped quantities, check scope, use ordinary scientific tools, verify units/direction/identity, validate a known case, inspect each mission, and preserve refusal when evidence is insufficient. E+ also received the actual retained module. [E5]

A common external scorer within the same team checked eight numerical missions and four appropriate scope/source refusals. All arms passed. The numerical cases include unfamiliar two- and three-dimensional inputs and affine maps; they remain a small authored engineering bank, not a population sample of microscopy customers. The result establishes zero measured accepted-quality difference for E versus U and E+ versus E on this bank. It does not imply that expert workflows never help.

Each arm used four underlying tool calls, or eight when orchestration wrappers are counted. Only U recorded a complete wall-clock duration (78.83 seconds); the other arms did not. Token counts, reasoning-token spend, invoices and complete customer/human effort were unavailable. Matched ceilings are equal authorized envelopes, not equal actual spend. Consequently no cost or latency advantage is inferred from the tie. U repaired a missing import in receipt generation; its measurement outputs were unchanged. [E5]

The cheapest next test of expertise

For a simple accepted-invoice model, let price be P, acceptance probability q, accepted-invoice default probability d, complete delivery expense a, and extra method expense Δa. A method with setup/retention cost B over N realized uses needs

$$\Delta q > \frac{\Delta a + B/N}{P(1-d)}$$

to improve expected contribution, holding price, quality meaning and settlement terms fixed. At assumed CAD425 and 4% default, CAD1 extra per job with no fixed cost requires about **0.245 percentage points** of acceptance improvement. The threshold must fit the available headroom $1 - q$. This simple calculation is a design aid; the fuller simulator includes partial delivery and refunds. A low-cost method should be evaluated only enough to resolve this economically meaningful difference. [E6]

7. Construct the offering and its arrangement together

The proposed offering is a source-bound quantitative transfer report: declared measurements, native and mapped results, exact inputs, validation checks, limitations, and machine-readable evidence. The narrow prototype does not deliver a full interoperable NGFF conversion, segmentation or validated registration. A prospective buyer must agree that this deliverable satisfies its actual scientific purpose.

CHUM's cellular-imaging service lists formatted image-analysis rates of $129.75 per hour for external academic/hospital clients and $143.50 per hour for businesses. Its stated scope includes staff analysis and quantitative extraction with tools such as FIJI, CellProfiler and Imaris. This is a relevant existing alternative, not a quote for the proposed report. CAD is an explicit inference from the Montreal provider; the retrieved rate block uses a dollar sign without spelling out currency. [8, 9]

At the business rate, assumed human durations of 1.5, 2.5 and 4 hours correspond to CAD215.25, CAD358.75 and CAD574.00. The range exposes the decisive uncertainty: the human reference can be cheaper than the model-selected CAD425 quote. Buyers may select that alternative or decline. Hours, buyer values, residual obligations and demand are assumptions. The study does not price the service as a fixed discount to human work or a markup on token cost. [E6]

The simulation gives matched enterprises the same eight prospective buyers, resources and choices. Buyers face service value, quote, effort, outside options, withdrawal, common qualification failure and default. The participation expression subtracts the full quoted price from expected delivered value. Because conditional settlement can later reduce payment, this is a **conservative maximum-price screening rule**, not exact risk-neutral expected utility. That behavioral assumption remains explicit; it was not corrected retrospectively to obtain a better result. [E6]

Route	Supplied construction and commercial choice
Established expert	Available analytical tools and ordinary delivery
Technical construction alone	A new common adapter, built before binding demand, with CAD240 assumed extra A cost
Coordination alone	Contingent participation using existing tools, with CAD35 extra coordination cost
Combined	Contingent participation plus the common adapter

No route is privileged as "the invention." E can choose the existing method or stop. The new adapter has zero primary quality or labor advantage. Shared demand and deposits can be considered, but participation is never inferred merely from common technical structure. Price, deposit fraction, minimum participation and the decision to operate were selected on 24 development worlds, then frozen before 96 fresh seeds across 12 scenarios, four routes and three retention arms: **13,824 histories**. [E6]

The selected route uses established tools, a CAD425 quote, no deposit and a minimum of one participant. This is a bounded grid optimum under supplied assumptions, not a market-clearing price or evidence of actual willingness to pay.

8. Conditional work and selective retention

The primary E route has mean simulated net collections of CAD1,160.94, A cash expense of CAD534.14 and owner opportunity cost of CAD29.94. Its settled cash contribution is CAD626.80, and economic surplus is CAD596.86. These totals include a **fictional** originating job paying CAD550 against CAD310 A cash and CAD20 opportunity cost, for CAD220 surplus. Subsequent work contributes CAD376.86. Actual revenue remains zero. [E6]

Frozen route, primary E	Mean simulated surplus (CAD)	Interpretation
Established expert	**596.86**	Best of the supplied routes
Technical construction alone	356.86	New construction costs exceed its zero primary benefit
Coordination alone	566.24	Coordination adds cost without enough benefit
Combined	356.24	No demonstrated technical/coordination complementarity

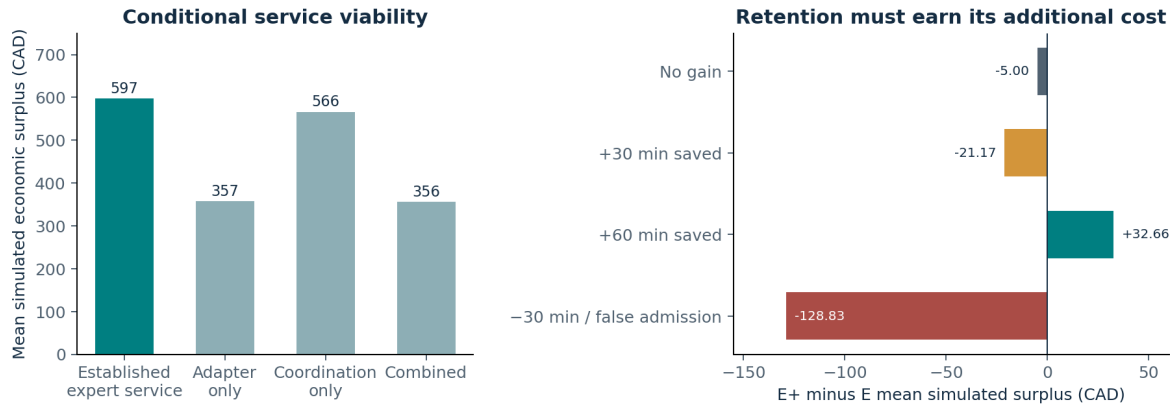


Figure 3: Conditional operating economics. Left: the established route outperforms the supplied construction and coordination routes. Right: incremental retention value includes zero, positive and negative-transfer assumptions. These are simulated outcomes; none is actual profit or measured human effort saving.

The primary E+ admission rule pays CAD5 to assess the opportunity, then rejects retention because no additional gain is supported. E+ earns CAD5 less in every primary paired world. This is a useful operating decision: profitable delivery, if later validated, need not depend on profitable sharing. The research artifact is retained for inspection and prospective tests; that does not make it an economically admitted Alpha asset.

An assumed half-hour saving per eligible use gives mean E+ minus E value of **−CAD21.17**. Forecasting four uses did not ensure their realization. An assumed one-hour saving gives **+CAD32.66**; a false admission with an actual half-hour penalty gives **−CAD128.83**. The same-content ordinary-library arm matches E+ exactly in every history. Neither a network premium nor automatic benefit from experience is embedded in the results. [E6]

Four of 96 primary E worlds lose economic value even with the fixed positive originating job. The interval for the simulated mean is CAD520.92–672.80; it describes Monte Carlo variation within assumptions. It is not evidence that those assumptions hold. Thinner demand, common failure, labor cost and buyer effort materially change results. A low-capital case stops before the origin, avoiding an unaffordable commitment. [E6]

9. Cash, costs and obligations stay distinct

The simulator records cents exactly and separates job delivery, extra retention and optional research. Original construction necessary for the originating paid outcome is A; it is not charged again when the artifact is retained. Unpaid research in this actual study is not relabeled as customer-funded learning.

Cost class	Included scope	Primary treatment
A: complete paid outcome	Intake/delivery labor, required construction, adaptation, failed attempts, corrections, support, tools, operating overhead and coordination needed for orders	CAD534.14 mean simulated cash expense, including fictional origin
B: additional reuse	Screening, rights/provenance packaging, admission, sharing and upkeep beyond A	E: zero; primary E+: CAD5 rejected screen; admitted package: CAD75 total
C: optional development	Extra practice or research beyond the paid outcome	CAD100 optional Gym candidate evaluated and rejected in primary policy
Opportunity cost	Owner time under assumed valuation	Separate from cash; CAD29.94 primary mean
Customer effort	Preparation and adoption burden in buyer choice	Not enterprise revenue or a free residual obligation

Refundable deposits increase cash and refund liability together; they do not immediately increase spendable funds. Only earned milestones release the appropriate amount. Delayed invoices cannot finance earlier construction. Default concerns the outstanding accepted balance and is not also charged as a refund. Withdrawals, common failure, partial delivery and support retain incurred costs and settle the applicable obligations. [E6]

For cash C_t, refund liability L_t, unpaid commitments O_t and reserve R_t,

$$\text{available}_t = C_t - L_t - O_t - R_t.$$

Seed capital is financing. The mean primary ending available balance of CAD1,976.80 includes CAD1,500 seed capital; it is not earned profit. All 13,824 histories satisfy the cent-exact cash identity, and the declared settlement horizon clears liabilities/receivables. Eight targeted accounting tests examine deposits, delayed collection, partial refunds, default, affordability, disjoint costs, parity and adverse histories. These are simulator implementation checks. [E6]

The 24-hour capacity limit covers fresh delivery, correction and support only. Fixed construction, coordination, retention and practice charges require outsourced or separately provisioned capacity. Their staffing and lead times were not modeled. The reported feasibility therefore does not carry over to a system where the same constrained staff must perform those activities within those hours. No such labor is assumed to be free.

The actual native ledger is separate. It records zero actual collections, unknown complete authoring/model costs, and null complete profit and available funds. Artificial positive admission and monetary entries exercise lifecycle/accounting branches only; they are labeled conformance fixtures, not corroborating economic observations. Token issuance, customer savings and hypothetical opportunity value are not entered as revenue. Applicable $AGIALPHA utility remains separate from operating profit. [E4, E6]

10. Retain selectively, then revalidate

The economic question is not whether an artifact is reusable in principle. It is whether this recipient gains enough eligible future benefit, beyond what it already possesses, to justify additional cost and risk. Here the competent recipient already has the mathematics. Numerical transfer succeeds, but extra quality and labor savings are not established.

With total incremental B cost $B = 75$ CAD, avoidable labor expense $w = 65$ CAD/hour and N realized eligible uses, the simple matched-quality threshold is

$$\Delta t > \frac{B}{wN}.$$

Four uses require 17.31 minutes each; two require 34.62 minutes; one requires 69.23 minutes. With no eligible use there is no finite saving threshold. The word **avoidable** matters: reduced salaried idle time is not automatically cash, and the same saved capacity cannot be counted again as wage savings and speculative future revenue. [E6]

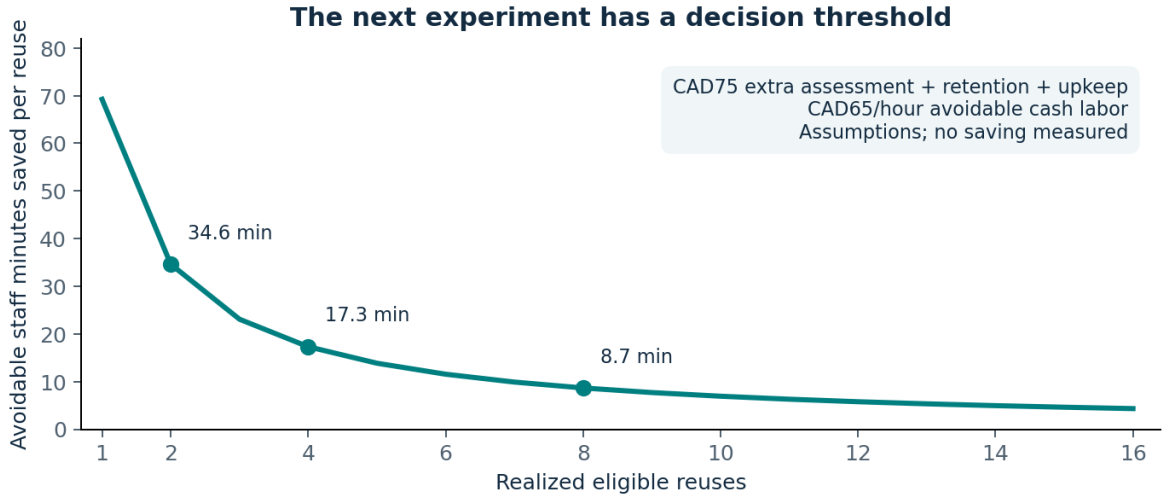


Figure 4: A conditional retention threshold. Each point assumes that many eligible uses actually occur and that saved time removes CAD65/hour of cash expense. Demand, negative transfer and additional costs can invalidate this simplified threshold. The experiment measured no human-effort saving.

A feasible admission screen compares expected benefit with achievable headroom before investing in a large assessment campaign. It includes the cheap rejected screen, false admissions, upkeep, scope drift and obsolescence. Forecast reuse is an uncertain random variable, including zero. The half-hour loss is retained as evidence that a favorable point estimate can still misallocate funds.

The capability record carries scope, prerequisites, version, source digest, dependencies, license/provenance, measurement evidence and limitations. The current module is a fixed analytical instrument, not a fine-tuned model or autonomous learning loop. No optional Gym gain was observed; the modeled Gym is a separate C decision. Future practice must retain its original artifact, environment, feedback, update rule and protected evaluation before claiming an improved version.

For a new mission, stable source qualification may be reused only when identity and scope remain valid. Input bytes, labels, physical units, map direction, requested meaning, acceptance and current authority must still be checked. An unfamiliar biological purpose, nonlinear map or changed dependency is a reason for a new qualification path, not permission to reuse an old ACCEPT.

11. Three foundations, one executable discipline

Transformational Intelligence: Learning to Change the Game separates the intent charter from inherited formulation and evaluation instrument. The present witness applies that distinction to a compulsory raster step while keeping the useful quantitative purpose fixed. Its causal example supplies a two-world model and active intervention; an unrestricted Bellman planner matches the result. This historical boundary is preserved: supplied formulas and strong ordinary alternatives remain visible here as well. [10, E9]

Invent the Economy: Transformational Intelligence for Creating Viable Markets treats technical capability, parties, prices, responsibilities, financing and rights as one candidate economic possibility. Its supplied four-buyer case has positive expected contribution but requires adequate prefunding; ordinary enumeration with identical options matches the constructor. The present factorial compares construction, coordination, their combination and an unrestricted established route, then models withdrawal, partial delivery, default, refunds and support. No historical synthetic market result is used as evidence of these buyers. [11, E9]

Transformational Autonomous Enterprises: SUCCESSOR Ω Harness permits inheritance of stable useful structure while requiring fresh mission meaning and separate authority. Its historical reuse, replay and direct specialist all pass their valid/invalid tasks; the direct specialist can be cheaper. That lesson appears directly in this study: transferred code works, but the competent analytical comparator and ordinary library also work. A successor's reuse cannot certify a changed scientific purpose. [12, E9]

Existing interfaces, bounded integration

The study freezes the ecosystem at commit `47c17f6aca6b1bb8779dd3e47d5ba5f8af3329f4` and reuses a 29-file unchanged dependency closure plus the existing research lifecycle adapter. It exercises the native signed Journal, TrustRegistry, CostLedger and research artifact transport. Real Ed25519/AES-GCM operations are used with labeled fictional identities; no live ENS authentication or production settlement occurs. [E4, E9]

All 32 local interface controls pass: source identity, authorized retrieval, unknown member, nested name, export rights, separate authority, revocation, changed mission, protected material, ciphertext/context/key tampering, signed accounting and incomplete costs. Sixteen pre-freeze input-boundary probes support implementation development. These counts are separate from scientific mission acceptance. [E4, E8]

Crucially, the unchanged production capsule scope **refuses the new microscopy component**. The study does not install a competing registry or expand protected permissions. A conformance fixture can show that all authorized direct `*.agent.agi.eth` members receive shareable bytes; it does not make those identities live. Jobs validator roles and settlement boundaries remain intact. Revocation prevents new mediated access; it cannot erase copies already retrieved. [E4]

This connects the foundations constructively: preserve purpose, evaluate a viable arrangement, inherit useful structure, and stop at the actual authority boundary.

12. A practical operating blueprint

The next operator can use the study without adopting a new registry, a new model or an oversized platform. The working unit is a mission and its evidence, with a separately reviewed reuse decision.

Stage	Concrete action	Acceptance or decision
1. Establish the useful outcome	Ask the beneficiary which quantity, source partition, frame, deadline, effort and price are acceptable	A signed-off mission specification; unresolved meaning produces HOLD
2. Check the existing alternative	Give the competent analyst standard tools and the same formulation	Compare complete accepted delivery, not an artificially weak raster route
3. Execute within scope	Bind source bytes; validate axes/units/map; measure; run independent checks	Required IDs/counts and numerical tolerances; explicit unsupported-scope refusal
4. Deliver and settle	Provide report, machine-readable output, provenance and limitations	Customer acceptance, agreed partial milestones, correction and collection recorded separately
5. Assess incremental retention	Count additional packaging, rights, admission, upkeep and rejected-screen cost once	Retain only when the recipient's expected extra benefit exceeds those costs and risk
6. Equip a fresh mission	Retrieve exact authorized bytes through the existing lifecycle, or ordinary authorized library	New mission meaning and execution authority checked separately
7. Reconcile and decide	Compare actual collections, obligations, cash and complete A/B/C costs	Continue, revise prospectively or stop; no speculative earnings fund commitments

For a research deployment, the minimum prerequisites are a supported integer label array, declared physical calibration, an explicit invertible affine map, the frozen Python dependencies and a person or authorized process able to resolve scientific meaning. A full NGFF reader is not required for the bounded NPY interface. A production integration would need its own qualified scope and release review.

The English/French demonstration lets a reader inspect actual source-region moments and vary a mathematical rotation/scale. Native volume remains fixed; mapped geometric volume changes under the displayed map. It also exposes the conditional retention threshold. Those browser controls recompute a mathematical scenario from archived moments; they do not rerun Python, authenticate a network member, settle a job or record a new experiment.

The operating order matters. A useful delivery may justify continued service even when retention is rejected. An artifact may deserve research preservation even when commercial admission fails. Conversely, a technically successful tool cannot borrow profitability from a favorable scenario, and a profitable scenario cannot grant production authority. The connected example therefore ends at a justified decision point rather than pretending every flywheel stage has succeeded.

13. Supporting diagnostics and preserved corrections

The development record is intentionally additive. It separates observed failures from risks identified by review and preserves the frozen instrument even when later audit improves the evaluator. [E8]

Observation or review finding	Action before or after freeze	What may be claimed
Hashing then measuring a mutable mapping could bind different bytes	Before freeze: own a bounded byte snapshot	Risk addressed by design review; the old race was not reproduced
NPZ, empty input and malformed headers could escape expected behavior	Before freeze: bounded NPY header/payload checks and structured refusal	Tested input boundaries; not universal parser security
Negative nonzero IDs were unnecessarily excluded	Before freeze: allow signed IDs within exact JSON range	Scope corrected to match supported label semantics
Diagnostic JSON serialization failed on NumPy boolean	Before freeze: convert to ordinary boolean; retain outputs	An actual development failure and repair
Black failed to import its installed dependency	Ruff formatting used where applicable	No Black pass claimed
Post-run comparator could mask hypothetical NaN through `max`	Preserve original evaluator; add finite/duplicate/shape checks; rescore stored outputs	All stored outputs finite; counts unchanged; four NaN probes refuse
Initial provenance receipt and later snapshot inventory differed	Preserve both; add inventory reconciliation	Existing study-page snapshots retained without inventing retrieval history
U's receipt-generation import failed	Worker repaired receipt code without changing measurements	Tool correction counted; no scored-answer repair

The primary technical outputs are therefore not favorable reruns substituted after a failure. The post-run comparator hardening leaves the instrument, tasks, source arrays and original outputs unchanged. Every result, contract, independent reference, process receipt, stdout and stderr is retained. A separate offline reconstruction re-decoded the original compressed chunks and reproduced all three exported array digests. [E2–E3]

The no-gain results are equally part of the record. Giving the formulation to the original conventional solver removes a quality advantage; identical library contents remove a graph-specific advantage; competent U succeeds without the extra rubric; the new adapter and coordination fail to beat established service in the primary scenario. These are useful causal separations, not reasons to discard the successfully executed instrument.

The economic admission rule is achievable under its stated threshold but can still fail through overestimated reuse or negative transfer. No retrospective "recovered profit" is credited after identifying those risks. Any revised buyer-choice rule, caching implementation, qualification cost or learning policy needs a newly frozen prospective comparison.

14. The next productive possibility

The most informative next step is **four real, unfamiliar imaging briefs** from willing laboratories or an imaging core, chosen to span different source layouts and genuinely unresolved measurement questions. The technical question is already sufficiently clear to avoid building more infrastructure first. The unresolved variable is the work required to establish meaning and complete an accepted handoff.

Before assignment, a beneficiary and evaluator should define useful quantities, tolerances, report content, deadline, maximum customer effort and acceptable price. Use a competent existing workflow E and E+ with the frozen artifact, balanced or randomized across briefs where feasible. Let both use standard libraries, caching and the same scientific clarification. Record preparation, provenance checking, adaptation, retries, correction, communication, verification, packaging, latency, model/tool usage and actual invoices or avoidable labor expense. Do not insert a fixed human-review allowance in place of observation.

The primary decision is complete incremental cost at matched accepted quality. If four eligible uses actually occur under the present CAD75/CAD65 assumptions, E+ must save more than 69.23 total avoidable minutes before an additional margin exists. A pilot may update that threshold if measured costs differ. Customer effort, late or partial delivery, unsupported tasks and refusals remain part of acceptance. A four-brief pilot estimates feasibility; it does not establish sustained specialist superiority.

A cheaper parallel check can test the frozen E rubric against U on fresh mission specifications with hidden semantic traps and an expert-quality oracle. Its purpose is to identify an acceptance or correction-cost difference large enough to pay for the method. Stop if both remain competent and the remaining uncertainty is commercially immaterial. No new paid API campaign is required merely to make the evaluation look ambitious.

Boundaries that could change the conclusion

This study covers affine transformations of supplied finite label partitions, not registration estimation, nonlinear deformation, topology, intensity conservation, biological validity or complete cells. It uses one development image and two held groups in one new image/timepoint. Hosted trials have one worker per arm, no randomization and incomplete cost telemetry. Customer decisions, prices, labor savings and collections are modeled; the conservative buyer screen and separately provisioned fixed-cost capacity are assumptions. Protected production microscopy scope remains unqualified. [E1–E6]

The productive opportunity is therefore a disciplined service test: determine whether a portable, source-bound measurement contract reduces real intake, correction or verification effort for recipients who do not already possess the same capability. If it does, retain selectively and test further transfer. If it does not, use the competent existing method and preserve the instrument as a reproducible research result. Both outcomes advance the mission without converting ambition into invented evidence.

15. Reproducibility, provenance and responsibility

The study lives in `studies/agi-alpha-microscopy-transfer-v1/` as an additive edition. It preserves prior papers and does not replace the existing runtime, registry, release owner or historical baselines. Coordination was initiated in repository issue #4850. The repository's ordinary publication qualification process governs any canonical deployment; a browser-readable source edition is distinct from a production-qualified application. [13, E9]

Reproduction map

Evidence	Entry point and retained record
Protocol and freezes	`protocols/PROTOCOL.md`, `source-freeze.json`, task commitment
Development	`run_development.py`; `development/development-summary.json`
Fresh execution	`run_confirmation.py`, `confirmation_reference.py`; frozen result directory
Post-run audit	`audit_confirmation.py`; preserved-output rescore and NaN checks
Native integration	`integration/run_integration.py`; 32-control summary and actual unknown-cost ledger
Hosted comparison	`hosted_trials/` task/answer/receipt files; `score_hosted_trials.py`
Economic study	`economics/simulate.py`, configuration, policy freeze, results and accounting tests
Public inputs	Original compressed chunks, metadata, source URLs, licenses and array hashes in `data/`
Visuals and reader	`make_figures.py`, this manuscript/LaTeX/PDF, bilingual demonstration

Recorded numerical runtime: Python 3.12.14, NumPy 2.3.2, SciPy 1.17.0; raw data decoding used Numcodecs 0.16.3. Integration used Node 24.22.1, Pydantic 2.11.7 and cryptography 50.0.2. The repository requests Node 22.17.1 for its web client; this local research replay is not full release qualification. Numerical results and hashes are reproducible targets; wall clocks, cryptographic nonces and hosted-model generations are not asserted to be byte-identical. [E1–E6]

Official model documentation was checked on 27 October 2026. It lists `gpt-6-astra` and `gpt-6.1-terra`; the study uses hosted same-model workers and does not establish a billed API transaction or immutable backend snapshot. Standard short-context API anchors are USD10/50 per million input/output tokens for Astra and USD1/5 for Terra. Token workload and currency conversion in the simulator are assumptions; these prices do not price the hosted research session. ChatGPT Pro is a subscription, not an API model identifier or automatic API entitlement. No proprietary weights were modified. [14–16, E10]

Blin et al. and Walther et al. retain credit for the public imagery and label data, redistributed under CC BY 4.0 with DOI, source URLs, metadata and documented extraction. The code retains the repository's applicable license and third-party notices. AI assistants assisted research, implementation, analysis, writing, translation and layout; execution logs and source records support the claims. Vincent Boucher's requested attribution is retained. No personal health record or private customer data was accessed.

This repository is a conceptual research prototype. AGI and superintelligence describe aspirations, not demonstrated general intelligence. Nothing in this edition establishes financial advice, actual sales, live identity authentication, production qualification or a completed profitable Flywheel. Its concrete contribution is an executed, transferable scientific instrument and a transparent decision boundary for the next useful work.

References and evidence index

Primary external and foundational sources

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2. OME. *OME-NGFF sample datasets*, including IDR0062A and IDR0052A. Catalog.
3. SimpleITK. *LabelShapeStatisticsImageFilter* and *Transforms and Resampling*. API; official notebook.
4. scikit-image. *skimage.measure*, region properties, spacing and caching. Documentation.
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14. OpenAI. *GPT-6 Astra*, current official model documentation. Accessed 27 October 2026. Documentation.
15. OpenAI. *GPT-6.1 Terra*, current official model documentation. Documentation.
16. OpenAI. *API pricing*, Standard short-context input/output anchors. Accessed 27 October 2026. Pricing.
17. GTFS. *Pathways examples*; OpenTripPlanner. *Accessibility*. GTFS; OTP.
18. US EPA/Sandia. *Water Network Tool for Resilience*. Overview.
19. NIST. *ThermoData Engine* and *Autonomous Formulation Lab*. TDE; AFL.

Executed evidence within this edition

E1 — Frozen purpose and realization. `protocols/PROTOCOL.md`, `initial_commitment.json`, `source-freeze.json`, `confirmation-task-commitment.json`; frozen `src/microscopy_transfer.py` and conventional reference.

E2 — Development and source data. `evidence/DATA_PROVENANCE.md`, `data/development/`, `data/confirmation/`, original chunks/manifests; `development/development-summary.json` and preserved outputs.

E3 — Fresh recipient execution. `confirmation/RESULTS.md`, `confirmation/provenance.json`, `confirmation/results/frozen-v1/summary.json`, contracts/oracles/process receipts and output digests; audit records.

E4 — Existing integration. `integration/README.md`, `source_manifest.json`, `results/frozen-v1/summary.json`, `native_actual_unknown.json`, separately labeled `native_synthetic.json` and cryptographic/lifecycle receipts.

E5 — Hosted same-model comparison. `hosted_trials/pretrial_commitment.json`, inputs, frozen rubric, U/E/Eplus answers and receipts, common `scoring.json`; `score_hosted_trials.py`.

E6 — Conditional economics. `economics/config.json`, `parameter_registry.json`, `simulate.py`, policy freeze, `results/results.json`, complete history CSV, example ledger, accounting tests, `README.md` and `RESULTS.md`.

E7 — Opportunity selection. `evidence/opportunity_dossier.json`, including all four candidates, prior-art sources, retrieval statuses, beneficiary, gap, mechanism, experiment and falsifier.

E8 — Development and correction history. `development/history.json`, `integration/development/REVIEW_HISTORY.md`, input-review versions, confirmation audit and archive inventory addendum.

E9 — Additive foundation integration. `integration/foundation_audit.md`, `foundation_sources.json`, source manifests and immutable upstream closure.

E10 — Model and service references. `evidence/model_economics_sources.md`, dated official identifier, capability, price and human-service source dossier.

Exact source anchors

Instrument SHA-256: `a04326ce0504bfe3ac290b6cb65ee6f089cc66a607f70b7cc7869099250c77db`.

Economic simulator SHA-256: `5d967ccd2dfd2d587b7e82d3bb61013d91e3f3c917aac381ffff08b2197c59e0`.

Economic policy freeze SHA-256: `2639f04946d2921b686ab460992893197e5a111fc9483d5f01881bbae1568f12`.

`source-results.json` and `results.tex` are generated directly from the retained result files; `build_paper.py` verifies the frozen instrument before compiling this edition. These authoring checks do not add new experimental observations.